

Concatenated Pattern in a Text

University stringology team is studying a particular language L over alphabet A . One of their tasks is to locate all occurrences of words of L in a given text T over A .

The alphabet A is the same for all subsequent notions and definitions, therefore it will not be explicitly mentioned most times.

Let $D = (W_0, W_1, \dots, W_{N-1})$ be a finite list of N ($2 \leq N$) non-empty words, so called dictionary D .

Let E be non-negative integer, which is smaller than the length of the shortest word in D .

Let $C: \{0, 1, \dots, N-1\} \times \{0, 1, \dots, N-1\} \rightarrow \{0, 1\}$ be a function which assigns value 0 or 1 to each pair of indices of words in D .

We say that the distance of a word Y from a word X is d , if d is the minimum possible number of edit operations which are subsequently applied on X and which transform X into Y . There are two edit operations:

Insert operation either inserts a character anywhere in the given word, or it prepends a character to the word, or it appends a character to the word.

Replace operation replaces a character by another character anywhere in the given word.

Let us denote by symbol $WE(j)$ the set of all words over A , whose edit distance from the word W_j ($0 \leq j < N$) is at most E .

Now, we specify language L . Each word WL in L is created by the following process.

First, word W_0 is written down. W_0 will be a prefix of WL .

Then, words from sets $WE(j)$ ($0 \leq j < N$) are appended, one after another.

A word from the set $WE(j)$ can be appended immediately after a word from the set $WE(i)$ whenever $C(i, j) = 1$.

There must be at least one word appended after the initial W_0 .

Note, when $C(i, j) = 0$ and $C(i, k) = 1$, and a word X is in both sets $WE(j)$ and $WE(k)$, then X can be appended immediately after any word of $WE(i)$, because $C(i, k) = 1$.

Example:

$D = (aaa, bbbb, ccccc)$

$C(0,0) = C(0,1) = C(1,2) = C(2,1) = C(2,2) = 1$. For other pairs (i, j) it holds, $C(i,j) = 0$

$E = 2$

Each word in L starts with aaa . It is followed by (possibly) modified word aaa or by a (possibly) modified word $bbbb$.

A (possibly) modified word aaa can be followed by a (possibly) modified word aaa or by a (possibly) modified word $bbbb$.

A (possibly) modified word $bbbb$ can be followed only by a (possibly) modified word $ccccc$.

A (possibly) modified word $ccccc$ can be followed by a (possibly) modified word $bbbb$ or by a (possibly) modified word $ccccc$.

Each modification of a word consists of at most 2 edit operations.

Let $T = ddaaabdbbbccddccaa$. We repeat T below with indices and spaces, for better readability.

Also, words of L which are present in T are indicated, dot denotes places of concatenations:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
d	d	a	a	a	b	b	d	b	b	c	c	c	d	d	c	c	a	a	
		a	a	a	b	b	d	b											(replace b in bbbb)
		a	a	a	b	b	d	b	b										(insert d in bbbb)
		a	a	a	b	b	d	b	b	c									(insert d and c in bbbb)
		a	a	a	b	b	d	b	b	c	c	c	c	d					(replace c in ccccc twice)
		a	a	a	b	b	d	b	b	c	c	c	c	d	d				(replace c in ccccc twice)
		a	a	a	b	b	d	b	b	c	c	c	c	d	d	c			(replace c in ccccc twice)
		a	a	a	b	b	d	b	b	c	c	c	c	d	d	c	c		(insert d and replace c in ccccc)

The task

Given an alphabet A , a dictionary D , value of E , a function C , and a text T , list all indexes of positions in T where ends a substring of T which is also a word of language L .

Input

The first input line contains three integers NA , N , E , where NA is the number of characters in alphabet A , N is the number of words in dictionary D , E is non-negative integer. Alphabet A consists of first NA small letters of English alphabet $\{a, b, c, \dots\}$.

Next there are N lines, specifying the list of words in dictionary D , each line contains one word.

Next, there are N lines specifying the function C . For each i ($0 \leq i < ND$), the i -th line starts with index i . After the index i , at the same line, there are listed in arbitrary order and separated by spaces, all indexes j , for which it holds $C(i, j) = 1$.

The last line of input contains the text T .

It holds, $2 \leq NA \leq 26$, $2 \leq N \leq 100$, $5 \leq |T| \leq 10^4$. The value of E is smaller than the length of the shortest word in D .

Output

The output contains one text line with a sequence of integers sorted in ascending order, separated by spaces, and representing the list of all indexes of positions in T where ends a substring of T which is also a word of language L.

It is guaranteed that T contains at least one word of L.

Example 1

Input

```
4 3 2
aaa
bbbb
cccc
0 1
1 2
2 1 2
ddaaabdbbcccddccaa
```

Output

```
8 9 10 13 14 15 16
```

Example 2

Input

```
4 3 1
aaa
bbbb
cccc
0 1 2
1
2 2
ddaaaccacccaaacccbcdddddaccccccbbb
```

Output

```
9 10 18 19 33 34
```

Example 3

Input

```
5 4 0
ddd
ccc
bbb
aaa
0 0 1 2 3
1 0 1 2 3
2 0 1 2 3
3 0 1 2 3
eedddcccaaaeeadddbbb
```

Output

```
7 10 20
```

Public data

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout and stderr is available to him/her.

[Link to the public data set](#)